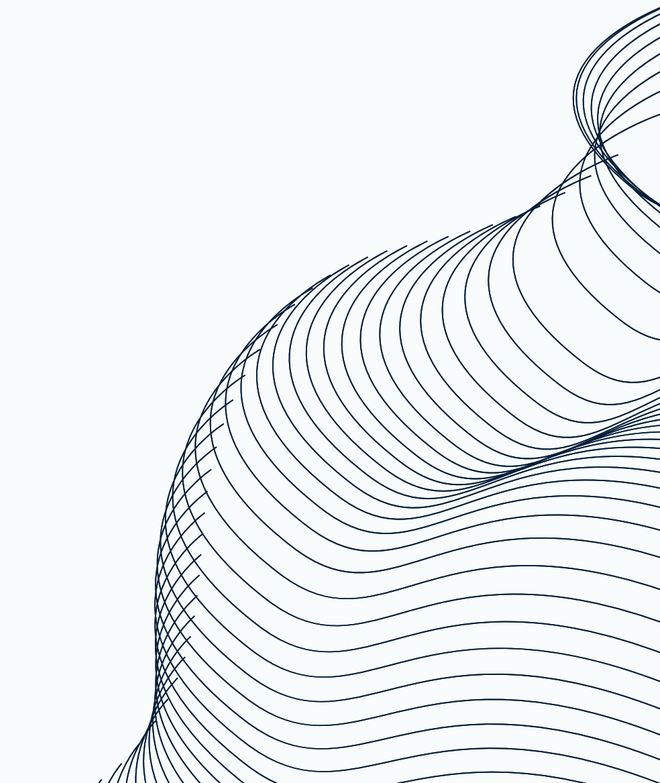
Alessandro Donina

Personal Portfolio



Updated to October 2026

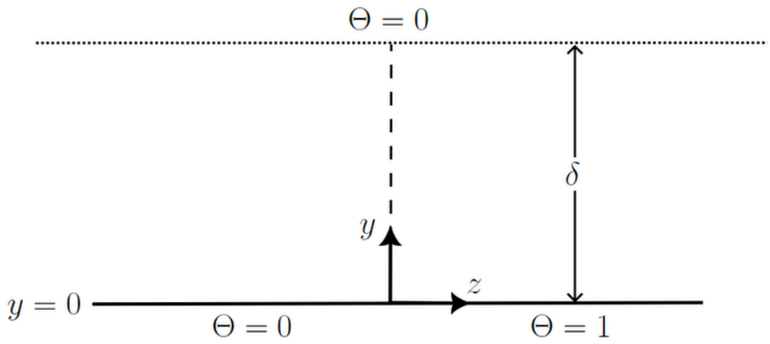
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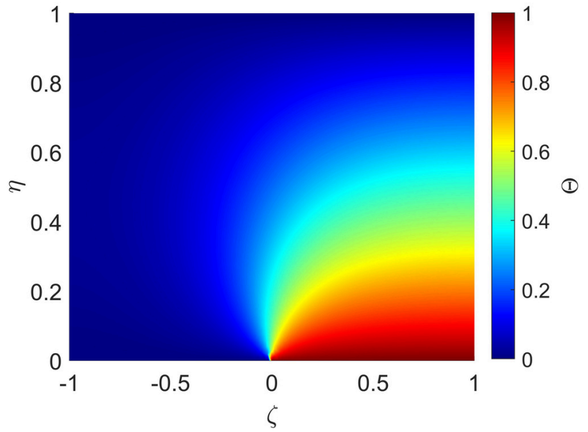
1. Bachelor’s Thesis

Solved thermal equation using a custom-made Matlab script

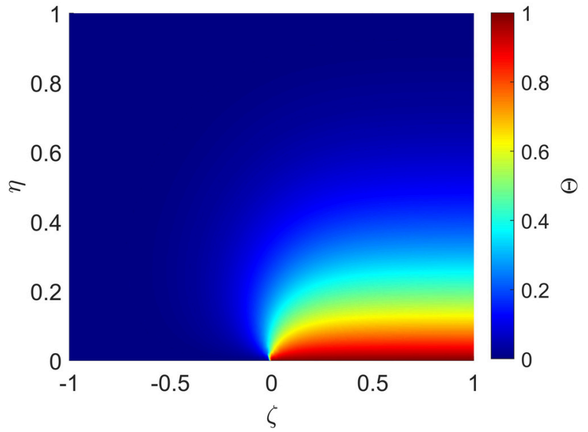
$$-\frac{Pr}{2}\left[f'\zeta\frac{\partial\Theta}{\partial\zeta}+f\frac{\partial\Theta}{\partial\eta}\right]=\frac{\partial^2\Theta}{\partial\eta^2}+\frac{\partial^2\Theta}{\partial\zeta^2}$$



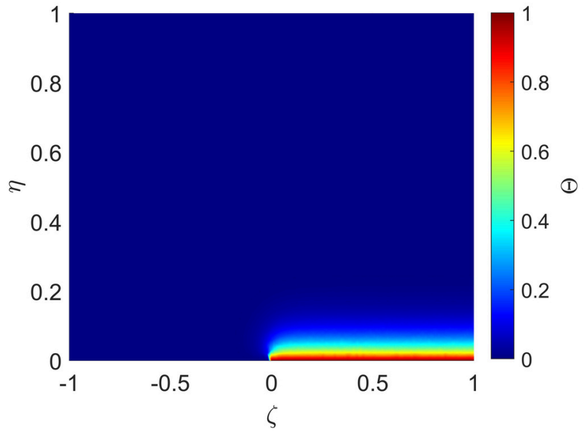
Problem Domain



Pr = 0.7



Pr = 7

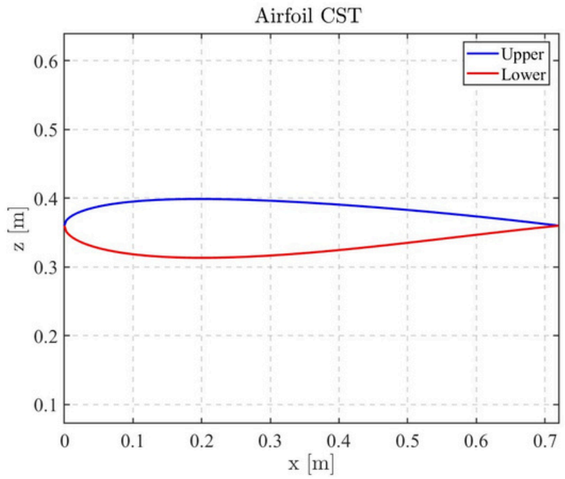


Pr = 50

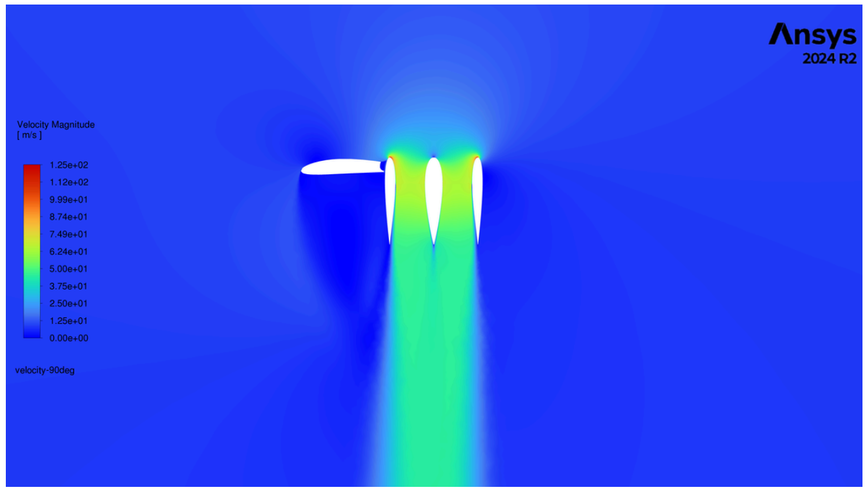
2. Ducted Propeller Design

Custom Bayesian optimizer for designing ducted propellers, validated through CFD.

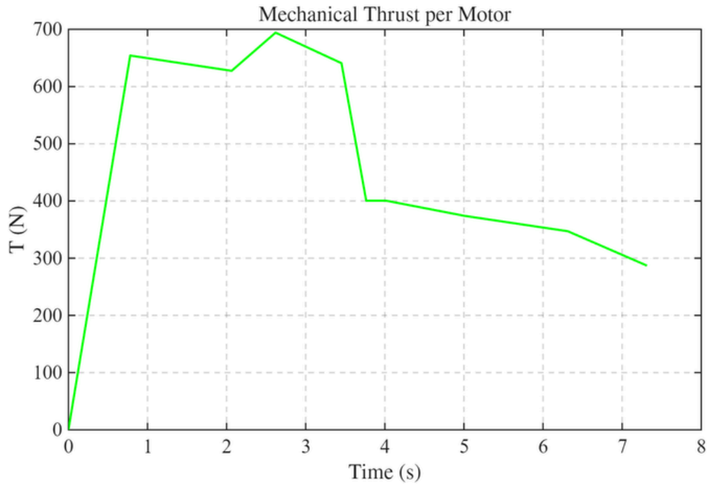
It generated up to 700 N of thrust with a 0.72 meter diameter duct.



Duct profile



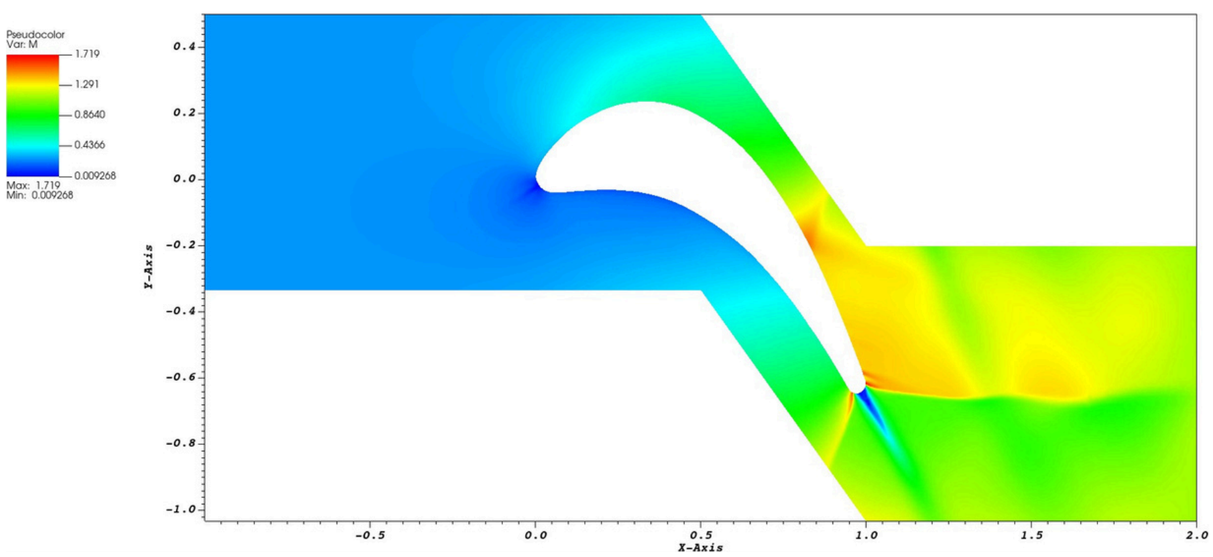
CFD Validation



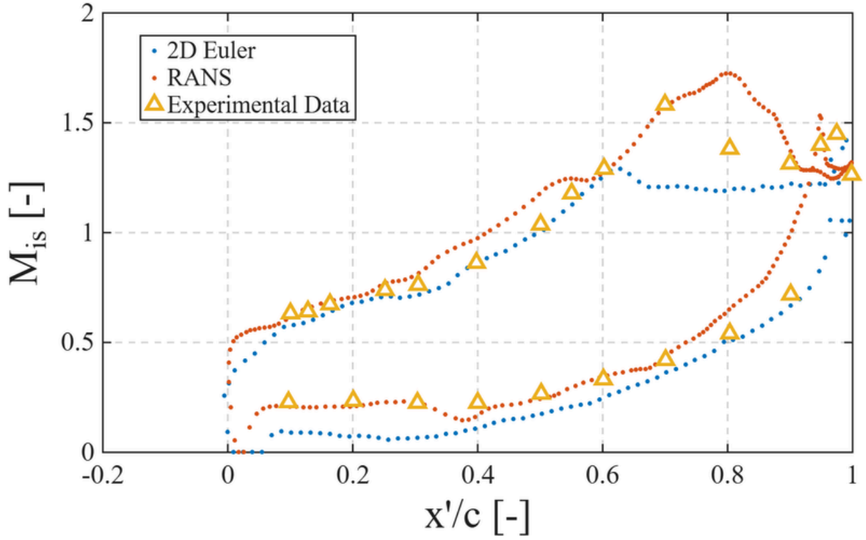
Trust

3. 2D Euler Code in Fortran

Developed a fluid dynamics code for the 'Computational Fluid Dynamics of Propulsive Systems' course, utilizing both the Roe and Lax-Friedrichs methods.



Test Case on LS59 Turbine Blade

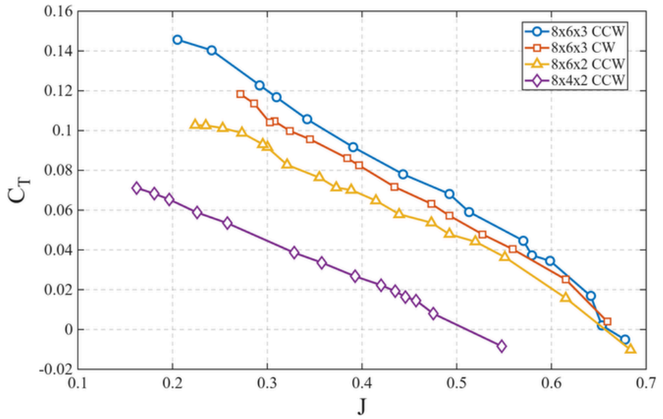


Comparison with RANS and experimental data

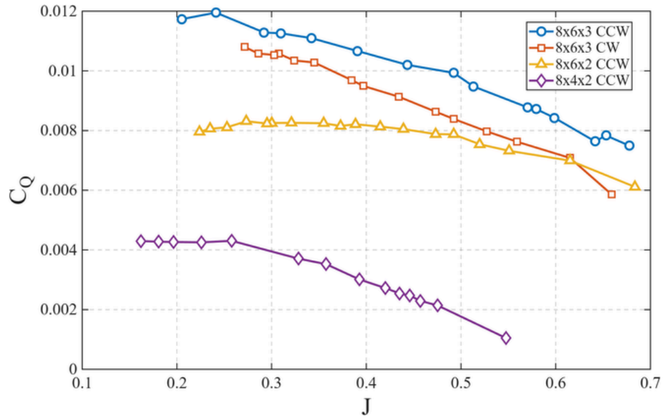
4. Propeller Performance Characterization

Developed a custom code to compute relevant quantities to characterizepropellers

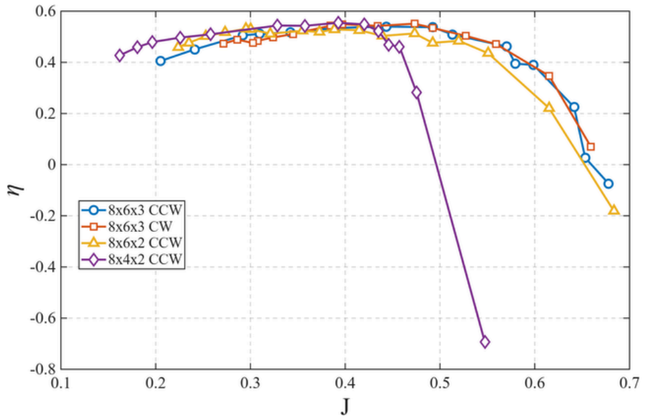
from raw data measured by a load-cell as the advance ratio changes.



Thrust Coefficient vs Advance Ratio



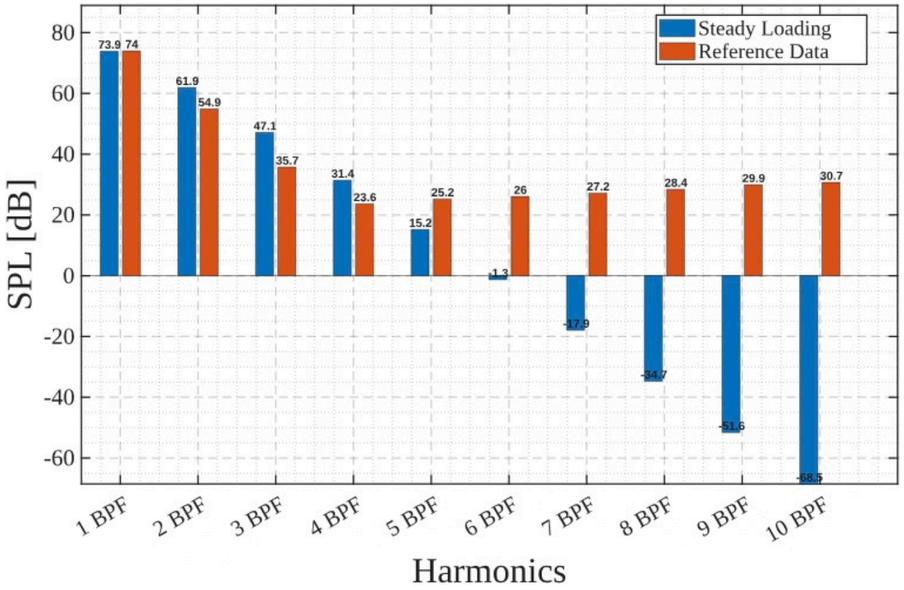
Torque Coefficient vs Advance Ratio



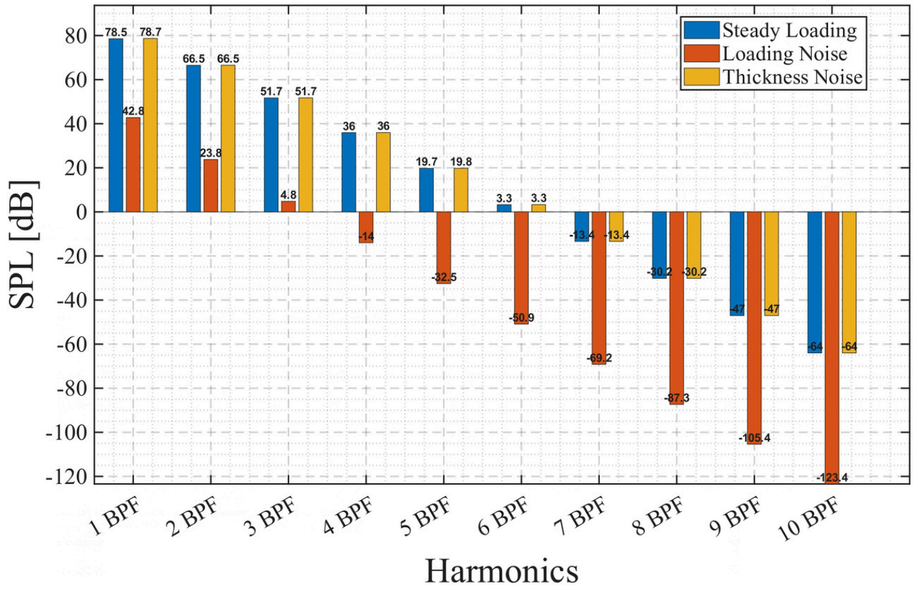
Efficiency vs Advance Ratio

5. Propeller Tonal Noise Analysis

Developed a custom code to compute propeller tonal noise in terms of both far-field spectra and directivity, in steady and unsteady conditions



Test case on a NACA 4412 propeller at 6000 RPM with axial velocity of 12 m/s*

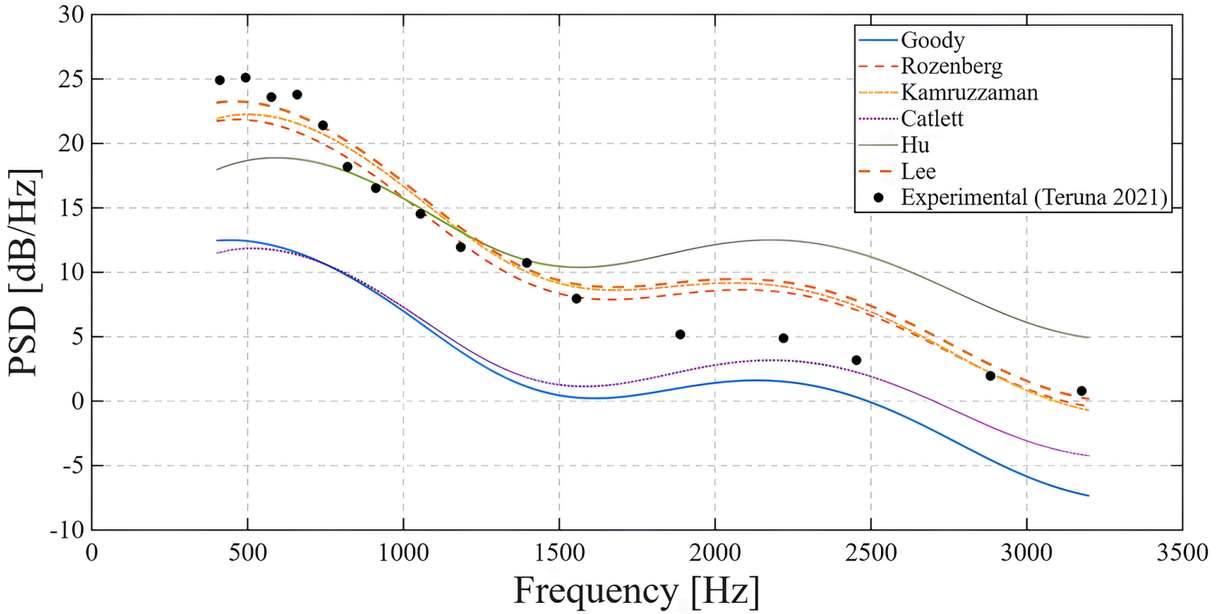


Test case on a NACA 3420 propeller at 6000 RPM with axial velocity of 16 m/s

* validated against high-fidelity simulations

6. Trailing Edge Noise Analysis

Developed a custom code to compute trailing edge noise using Amiet's and Roger’s diffraction theories, validating the results against experimental data.



Test case on a NACA 0018 comparing experimental data with numerical results across different wall-pressure models.

Thank you for your time!